Session 1: Math Prerequisites

KB8029 — Bioinformatics & Deep Learning

Arne Elofsson

SciLifeLab

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Log in to the lecture quiz

Log in to the quiz

Scan the code, or go to quiz.bioinfo.se/live

Log in with your university account

Answering counts as being at the lecture: up to 3 bonus points on the exam

Right or wrong is never graded, only whether you answered. Results are shown without names.

quiz.bioinfo.se/live

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Warm-up: what do you know already?

3 quick questions

On the topics of today, one at a time, about 30 seconds each

Answer on your own, and guess if you do not know

It shows me where to start; right or wrong is never graded

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A is a 3 × 2 matrix and v is a vector of 2 numbers. What is A v?

A A vector of 3 numbers

B A vector of 2 numbers

C A 3 × 2 matrix

D Not defined

Answer now
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What is the derivative of e^x?

A $x\, e^{x-1}$

B e^x

C $\ln x$

D $1/x$

Answer now
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What is $\log_2 8$?

A 3

B 4

C 2.08

D 16

Answer now
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What the room knows already

Where we start

Which topics most of you know, and which we take slowly today.

Show the results ▸ quiz.bioinfo.se/live/report

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In up to three words: what do you want to learn today?

Vectors and matrices

$x \cdot w$, $\|w\|$, Xw

Derivatives and gradient descent

$\partial L/\partial w$, ∇L, η

Probability and statistics

$P(A|B)$, log-odds, p

Answer now

Scan the code, or go to
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What you want to learn today

What the room wants to learn

Show the results ▸ quiz.bioinfo.se/live/report

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The mathematical vocabulary of the course

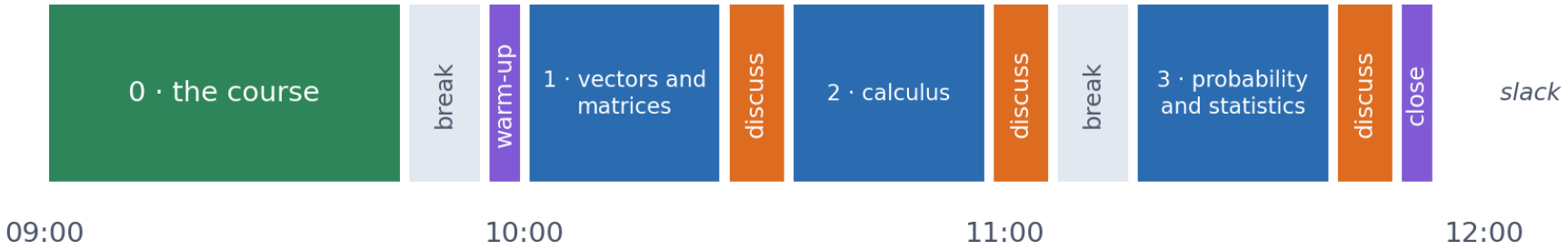
Vectors and matrices	Calculus	Probability and statistics
profile scores (S5)	gradient descent (S7)	log-odds: BLOSUM, PSSM (S4-5)
perceptron (S7)	backpropagation (S9)	cross-entropy loss (S8-9)
every layer (S7-12)		maximum likelihood: training

today's maths, and where it comes back

Not all the mathematics you might need: the ideas this course uses, worked through with real numbers, each with a pointer to where it returns.

See course.bioinfo.se/session01.html. This session has no lab; the first lab is in Session 2.

Three blocks: about 25 min talk and 8 min discussion



Block 1: vectors and matrices. Block 2: derivatives, gradients and gradient descent. Block 3: probability, likelihood, entropy and p-values.

Sessions 0 and 1 share one morning, 09:00-12:00, so Session 1 has about two hours, not three. Times are approximate: a 5 min start (warm-up quiz and your questions), one 10 min break and a 5 min close. Each block ends with a quiz question: guess, run the notebook and work out why it behaves as it does, discuss, answer again.

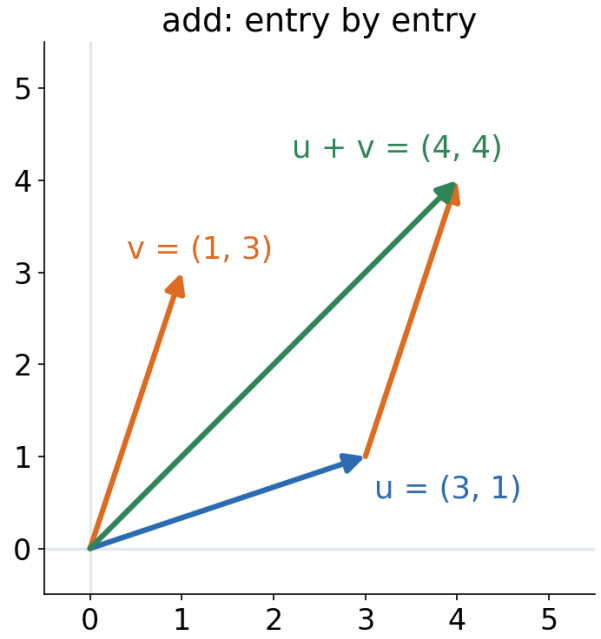
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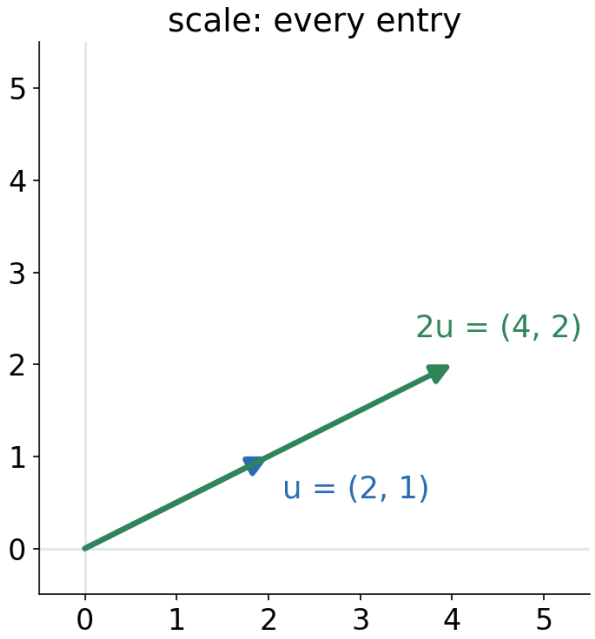
BLOCK 1 · ABOUT 25 MIN

Vectors and matrices

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A vector is an ordered list of numbers

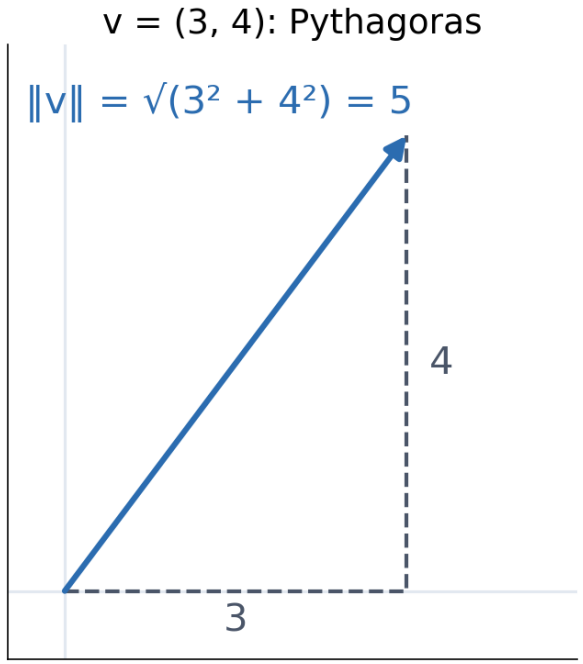




- **Add: entry by entry**
- **Scale: every entry**
- Dimension = number of entries

See course.bioinfo.se/session01.html, 'Vectors' and 'Dimensions and shapes'.

The length of a vector

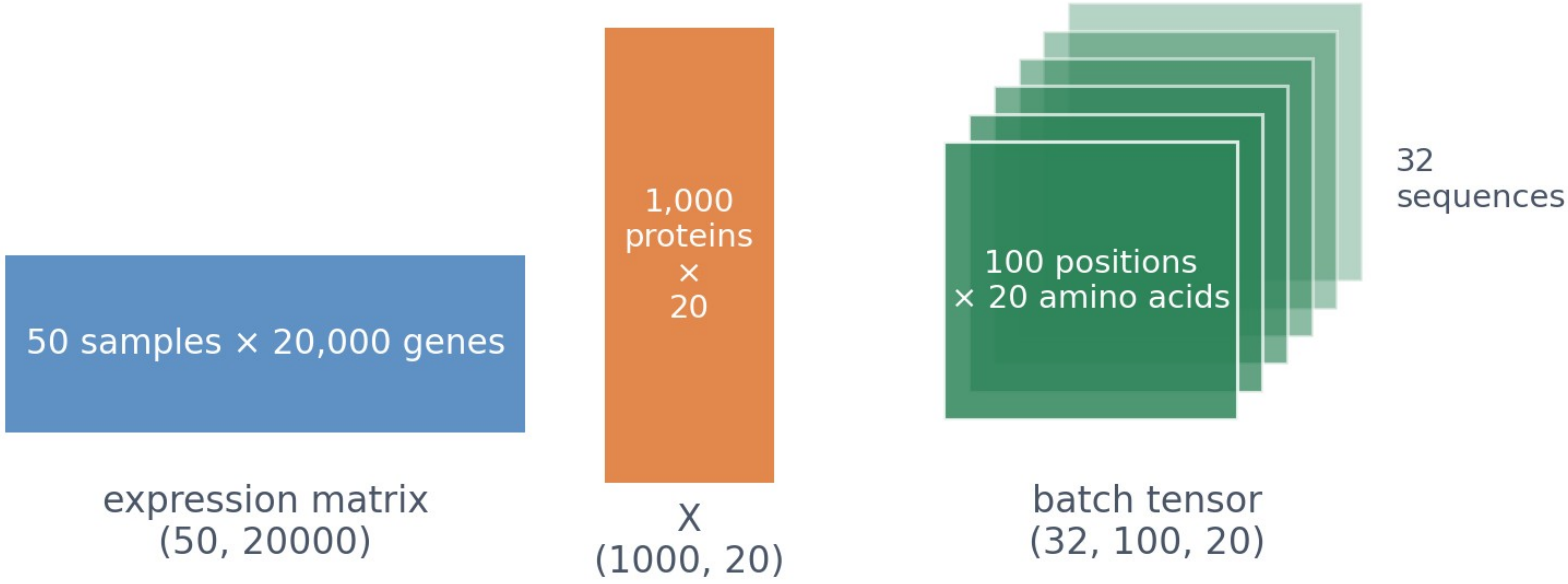


- $\|v\| = \sqrt{(v_1^2 + \ldots + v_n^2)}$
- Pythagoras in n dimensions
- Also called the norm

See course.bioinfo.se/session01.html, 'Vectors'.

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Always know the shape of every matrix

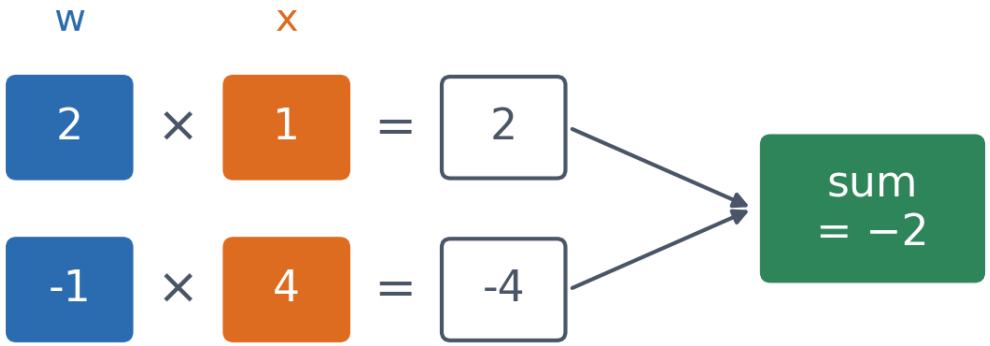


A matrix with n rows and m columns has shape (n, m). In biology a matrix is a data table: one row per sample or protein, one column per feature. A tensor continues to three or more dimensions.

See course.bioinfo.se/session01.html, 'Dimensions and shapes'. Tensors properly: Session 2.

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The dot product: multiply, then add



multiply the entries pairwise, then add: a weighted sum

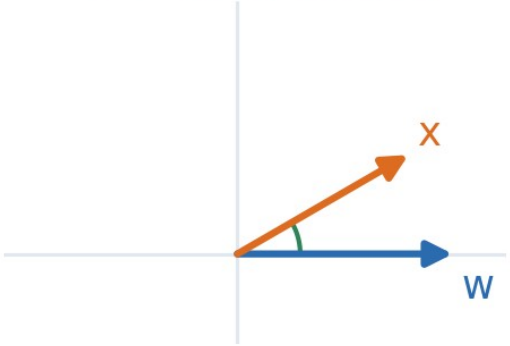
- $\mathbf{w} \cdot \mathbf{x} = w_1x_1 + \ldots + w_nx_n$
- A weighted sum
- $(2, -1) \cdot (1, 4) = -2$

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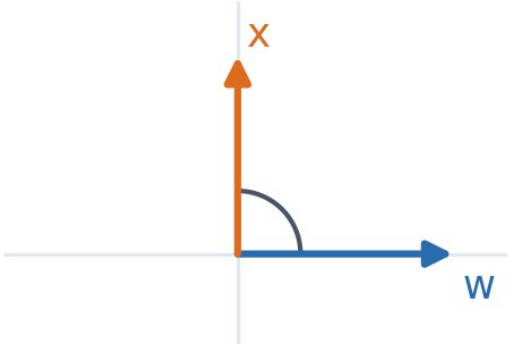
The geometric meaning: the angle gives the sign

$$w \cdot x = \|w\| \, \|x\| \cos \theta$$: the angle gives the sign

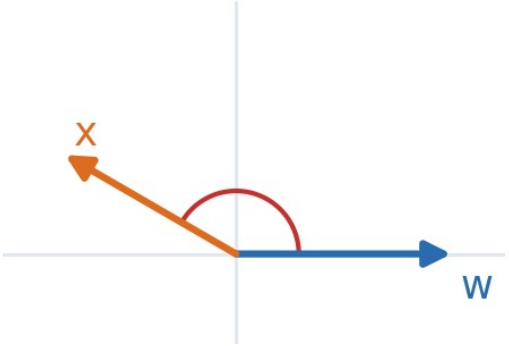
θ < 90°: positive



θ = 90°: zero



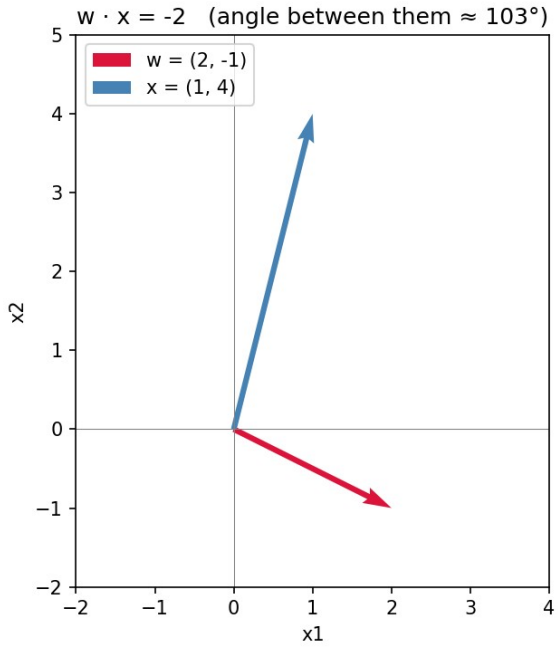
θ > 90°: negative



Positive when the vectors point broadly the same way, zero when they are perpendicular, negative when they point broadly opposite ways.

See course.bioinfo.se/session01.html, 'The dot product'.

(2, −1) and (1, 4) are about 103° apart

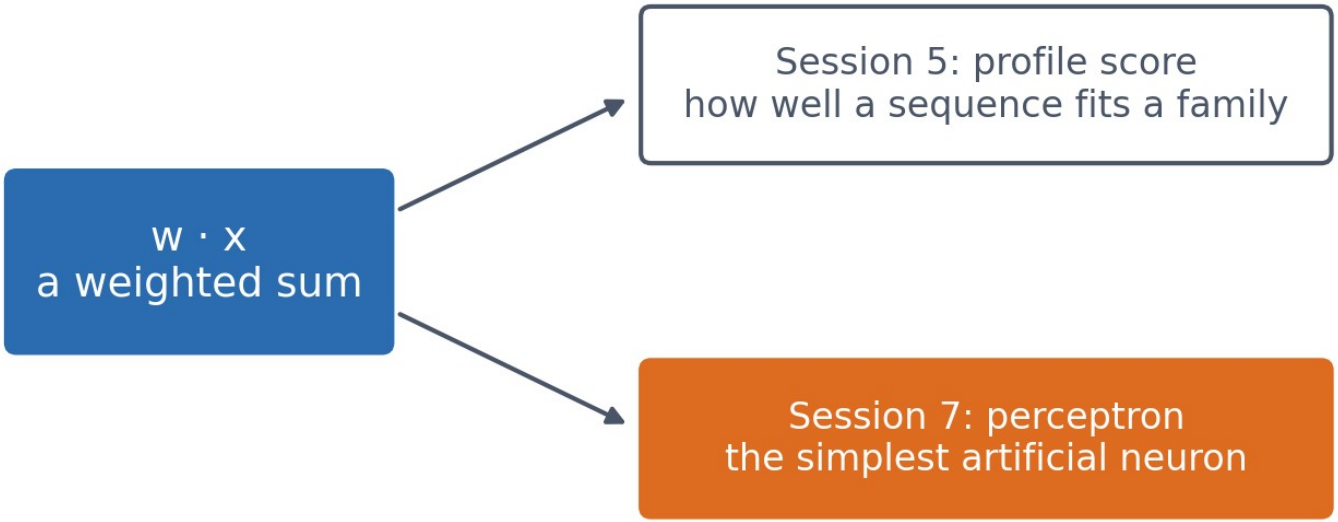


- Just past perpendicular
- **So w · x < 0**
- No arithmetic needed for the sign

See course.bioinfo.se/session01.html, 'The dot product'.

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Why it matters: one weighted sum, twice

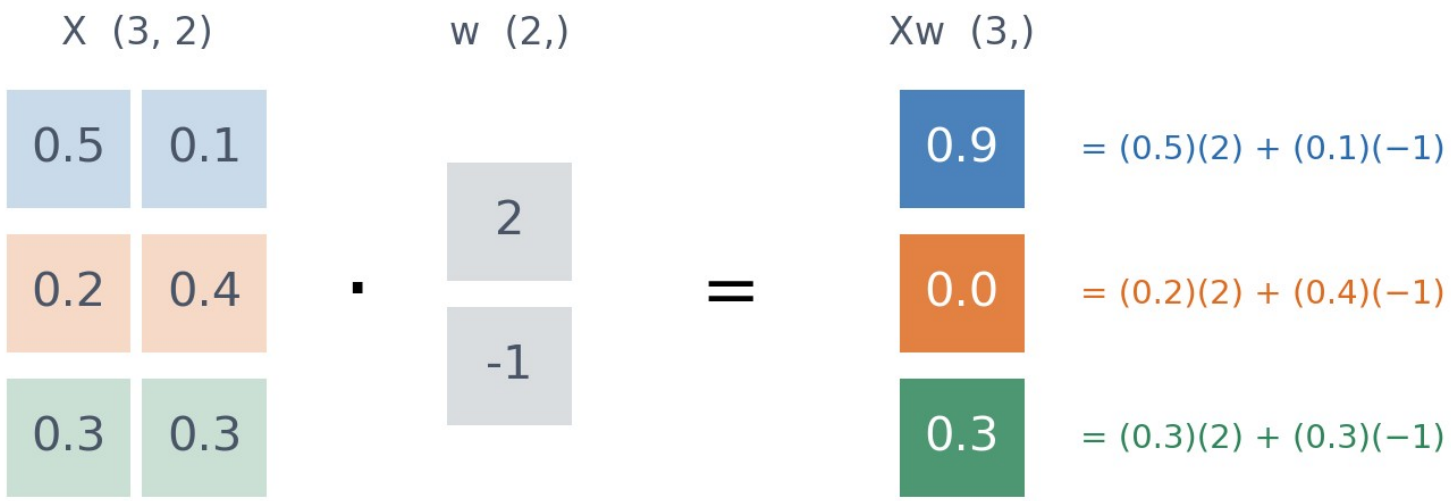


A profile score multiplies each position's value by a position-specific weight and adds them up. A perceptron computes the same dot product. When you get there, only the interpretation is new.

See course.bioinfo.se/session01.html, 'The dot product'.

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A matrix-vector product is one dot product per row

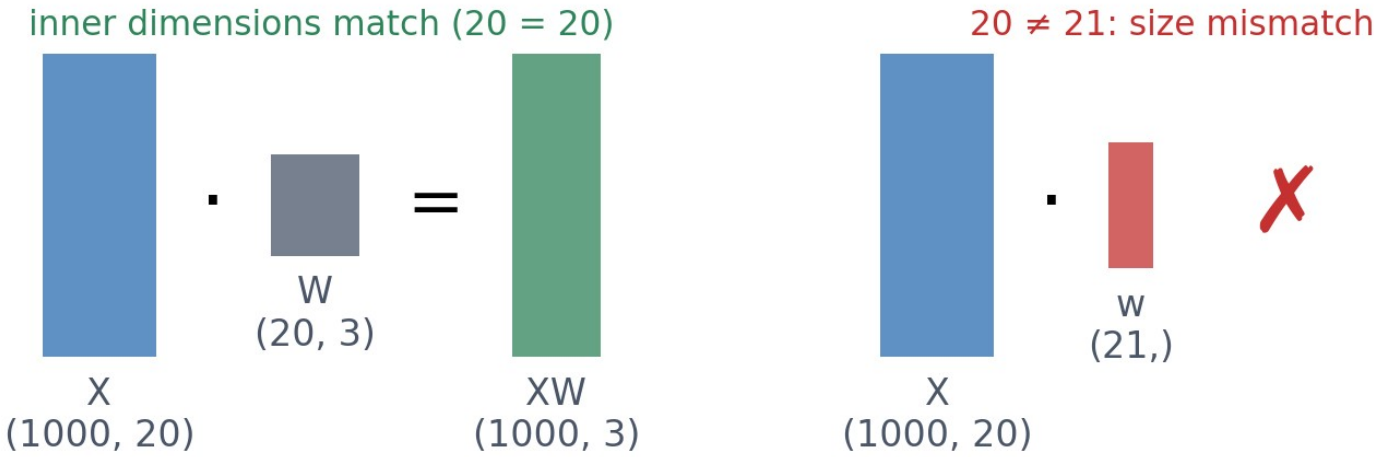


Three proteins, two features (hydrophobic and charged fractions), w = (2, −1). One product gives all three scores, with no loop: this is how classifiers score every example at once.

See course.bioinfo.se/session01.html, 'Matrix multiplication'.

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Inner dimensions must match

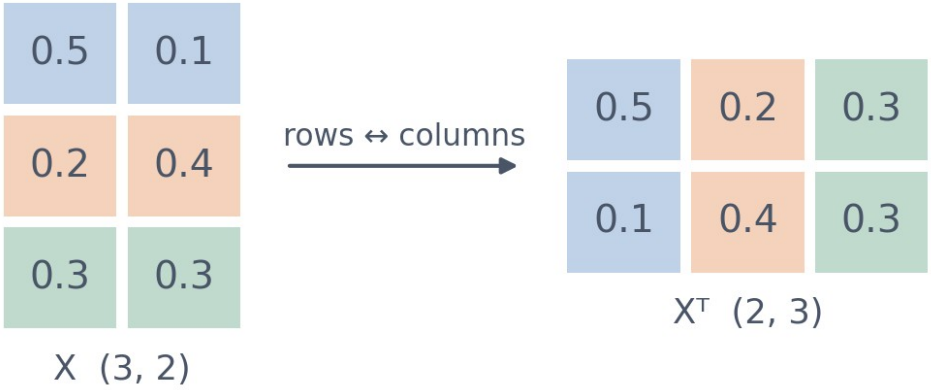


$$(n \times m)(m \times k) \rightarrow (n \times k)$$

Three classes: W is (20, 3), and XW gives three scores per protein. A weight vector of length 21 cannot be multiplied: the most common error in this course's code.

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The transpose swaps rows and columns

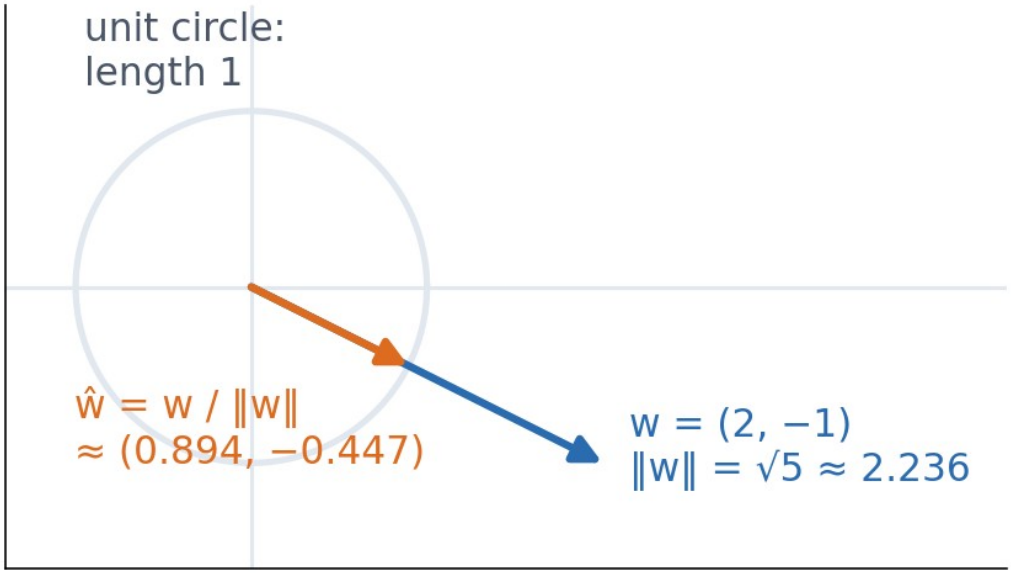


- **$w^T x$: the dot product**
- $X^T X$: a (20, 20) matrix
- $(AB)^T = B^T A^T$

$X^T X$: (20, 1000)(1000, 20) → (20, 20), every feature against every feature

See course.bioinfo.se/session01.html, 'Matrix multiplication'.

Normalising keeps only the direction



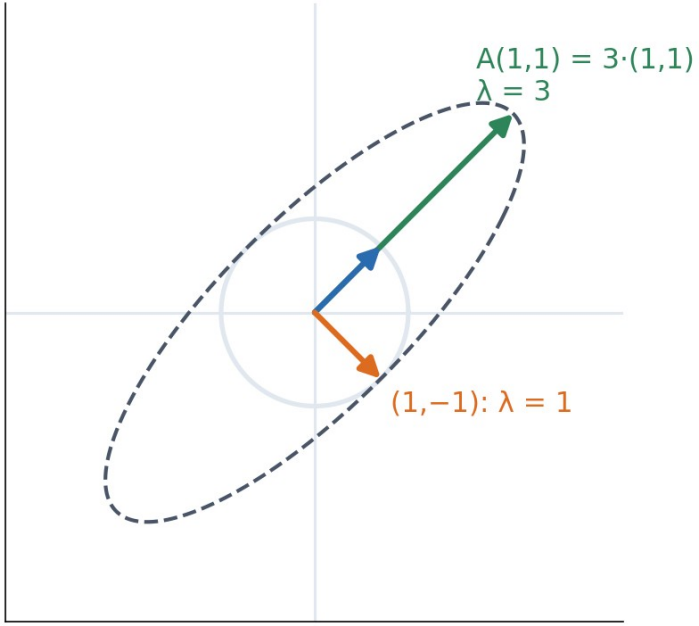
- **$\hat{w} = w / \|w\|$**
- Length exactly 1
- Unit · unit = cosine similarity

See course.bioinfo.se/session01.html, 'Normalisation'.

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Eigenvectors keep their direction

A = [[2, 1], [1, 2]] turns the circle into an ellipse



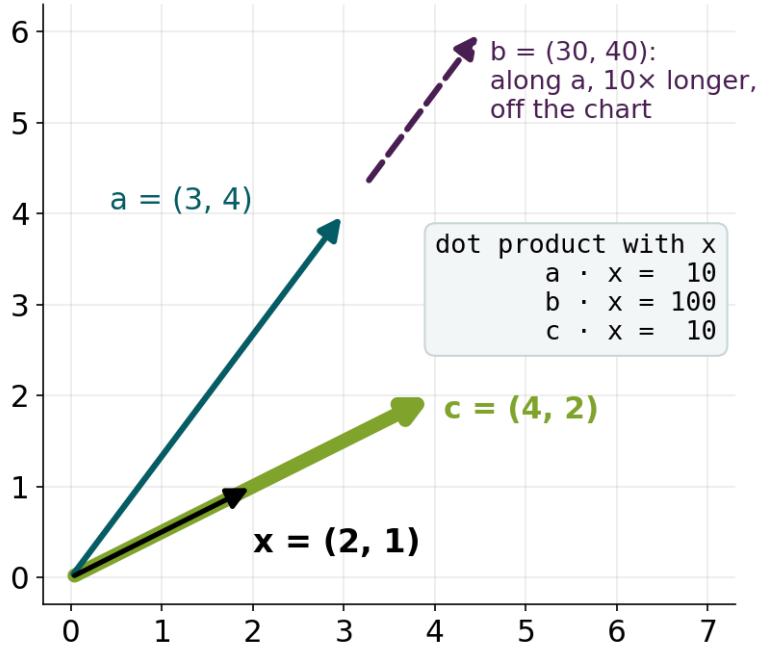
- **Av = λv**
- Check, don't solve
- PCA: eigenvectors of the covariance matrix

See course.bioinfo.se/session01.html, 'Eigenvectors and PCA, intuitively'.

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Which vector points most in the direction of x?

Smallest angle with x = (2, 1). The figure shows each vector and its dot product with x.



A b: ten times larger dot product

B a and b equally: same direction

C c: points exactly along x

D a and c equally: same dot product

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

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Which vector points most in the direction of x?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

SciLifeLab

Which vector points most in the direction of x?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session01-discussion-1.ipynb

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Which vector points most in the direction of x?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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Which vector points most in the direction of x?

Results, and who explains

Both rounds side by side.

Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 1 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session01-discussion-1.ipynb

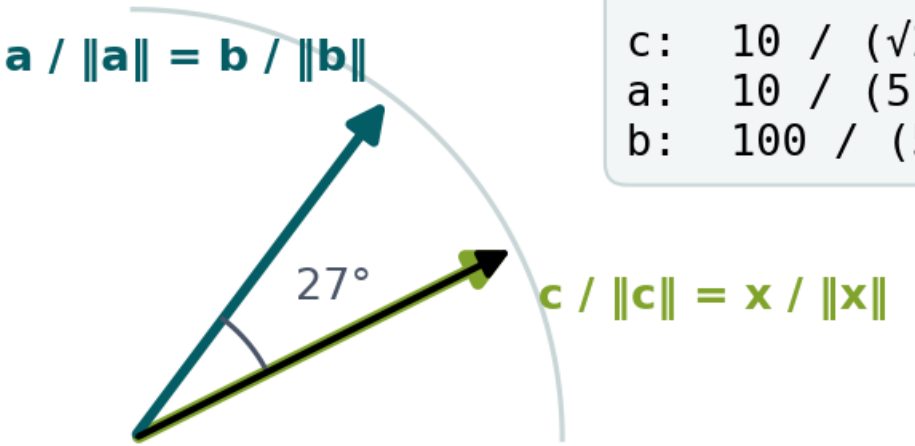
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The answer: c points along x

Normalised: length 1 each, only the direction is left

$$\cos\theta = w \cdot x \,/\, (\|w\|\,\|x\|)$$

c:	$10 / (\sqrt{20} \cdot \sqrt{5})$	$= 1.00$	$(0°)$
a:	$10 / (5 \cdot \sqrt{5})$	$= 0.89$	$(27°)$
b:	$100 / (50 \cdot \sqrt{5})$	$= 0.89$	$(27°)$



Normalising (dividing by the length) leaves only the direction. b is a made ten times longer, so its dot product is ten times larger at the same angle; a and c have the same dot product but point differently. The dot product mixes length and direction: $w \cdot x = \|w\|\,\|x\| \cos\theta$.

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BLOCK 1 DONE

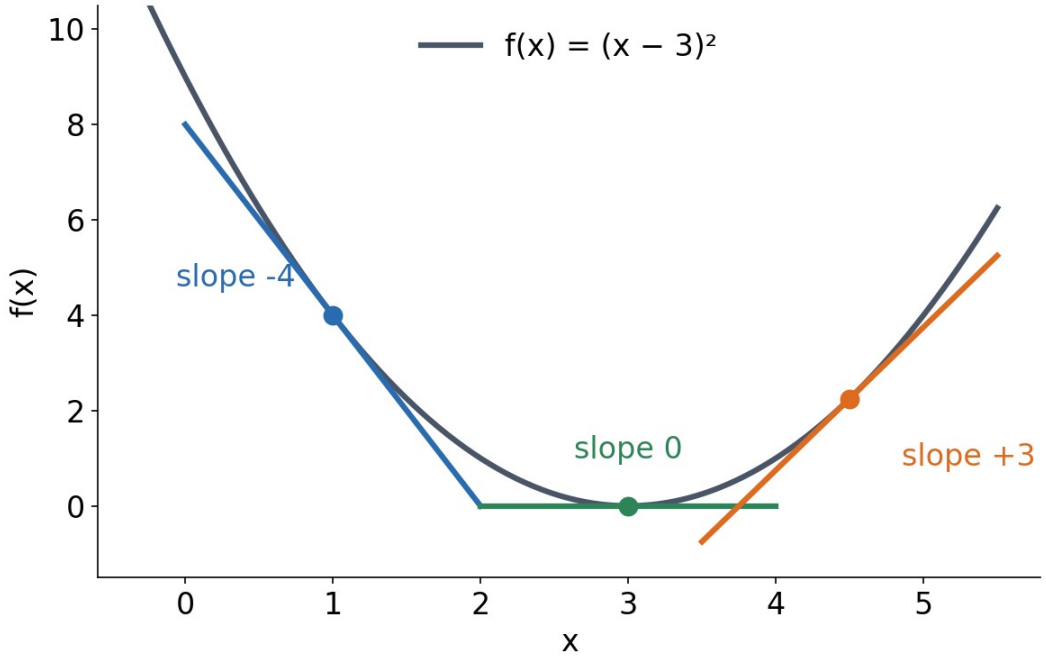
Break

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BLOCK 2 • ABOUT 25 MIN

Calculus for machine learning

A derivative is the slope of the tangent



- **How fast the output changes**
- Zero at the minimum
- $f'(x) = 2(x - 3)$ here

See course.bioinfo.se/session01.html, 'Derivatives'.

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The derivatives used throughout the course

f(x)	f′(x)
constant c	0
x^n	$n\,x^{n-1}$
e^x	e^x
ln(x)	1/x
sin(x)	cos(x)
cos(x)	−sin(x)
$\sigma(z) = 1/(1 + e^{-z})$	$\sigma(z)(1 - \sigma(z))$

- Worth memorising
- **The sigmoid's derivative uses σ itself**
- Backpropagation needs it (Session 9)

See course.bioinfo.se/session01.html, Table 1.1.

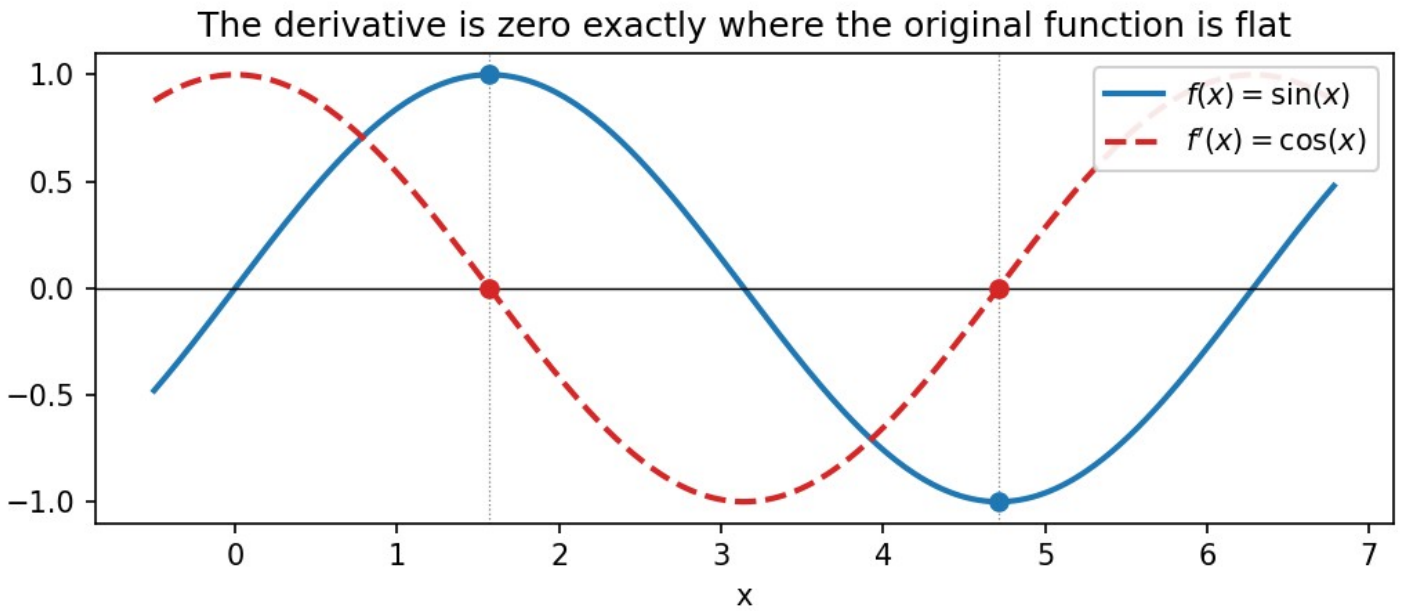
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One row matters more than the others

- $\sigma(z) = 1/(1 + e^{-z}) \rightarrow \sigma'(z) = \sigma(z)(1 - \sigma(z))$
- Written in $\sigma(z)$, not in e^{-z}
- $\sigma(z)$ is already computed on the way forward
- **So the derivative costs nothing extra**
- Session 9 needs it for every sigmoid unit

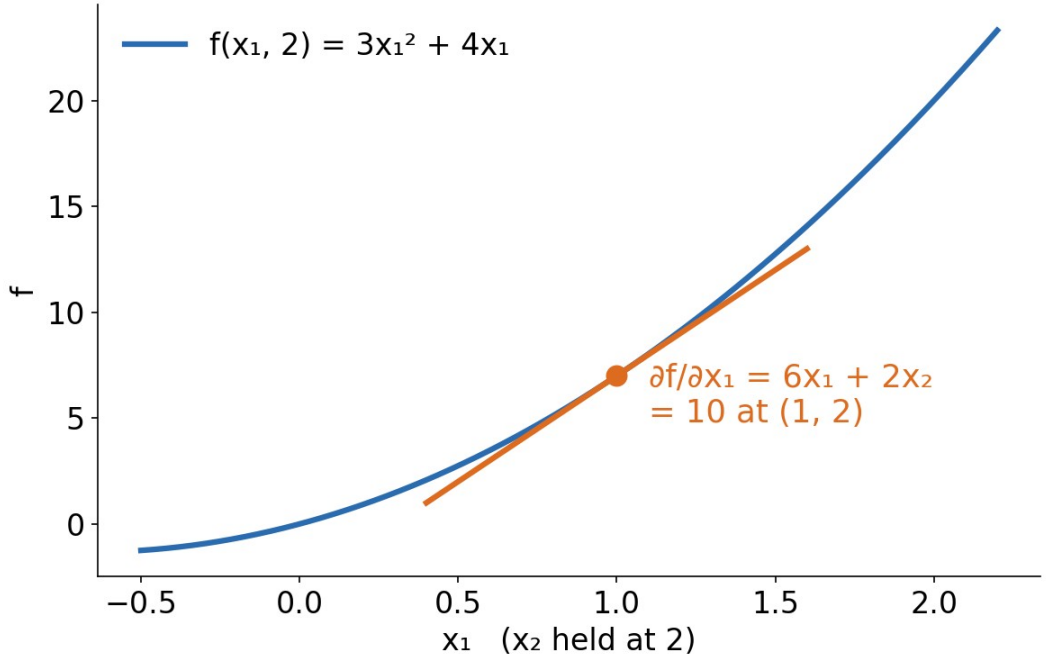
Table 2 on course.bioinfo.se/session01.html.

Peaks and troughs: where the derivative is zero



f(x) = sin(x) (solid) and f′(x) = cos(x) (dashed). Wherever sin(x) has a peak or a trough, cos(x) crosses zero.

A partial derivative: hold the others fixed



- $\mathbf{f(x_1, x_2) = 3x_1^2 + 2x_1x_2}$
- $\partial f/\partial x_1 = 6x_1 + 2x_2$
- $\partial f/\partial x_2 = 2x_1$

See course.bioinfo.se/session01.html, 'Partial derivatives and the gradient'.

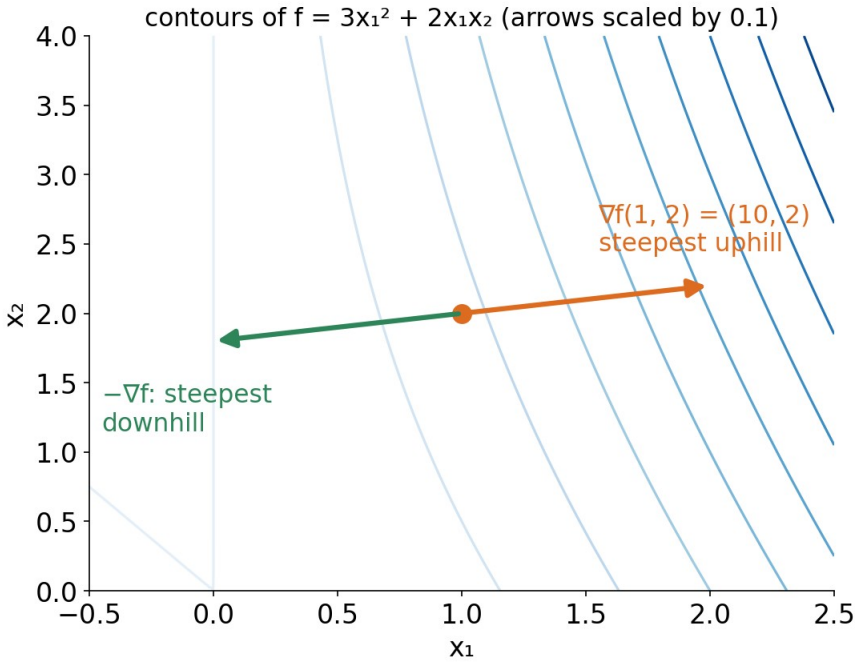
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A worked gradient, in full

- **$f(x_1, x_2) = 3x_1^2 + 2x_1x_2$**
- $\partial f/\partial x_1 = 6x_1 + 2x_2$ (x_2 held constant)
- $\partial f/\partial x_2 = 2x_1$ ($3x_1^2$ contributes nothing)
- **$\nabla f(1, 2) = (10, 2)$**

See course.bioinfo.se/session01.html, 'Partial derivatives and the gradient'.

The gradient points uphill

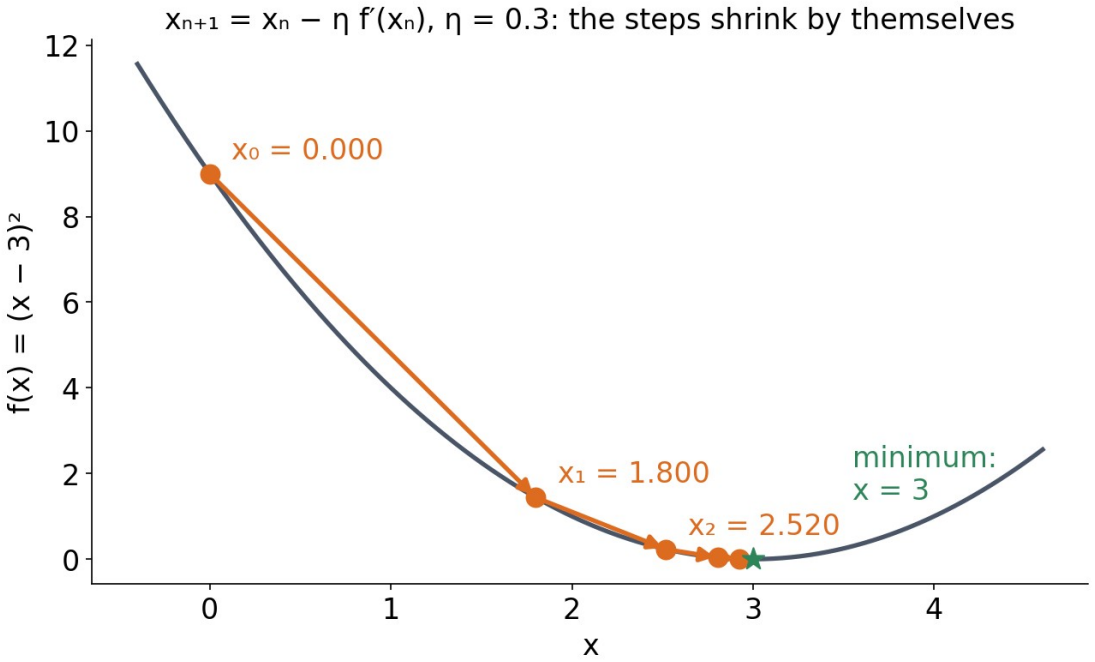


- **$\nabla f = (\partial f/\partial x_1, \ldots, \partial f/\partial x_n)$**
- $\nabla f(1, 2) = (10, 2)$
- Direction: steepest uphill
- Length: how steep

See course.bioinfo.se/session01.html, 'Partial derivatives and the gradient'.

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Gradient descent, one step at a time



- $x_{n+1} = x_n - \eta\, f'(x_n)$
- 0 → 1.800 → 2.520 → 2.808
- Closes in on x = 3
- The steps shrink by themselves

See course.bioinfo.se/session01.html, Table 1.2.

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The same four steps, as arithmetic

- **$f(x) = (x - 3)^2$, $f'(x) = 2(x - 3)$, $x_0 = 0$, $\eta = 0.3$**
- n = 0: x = 0.000, $f' = -6.00$ → 1.800
- n = 1: x = 1.800, $f' = -2.40$ → 2.520
- n = 2: x = 2.520, $f' = -0.96$ → 2.808
- n = 3: x = 2.808, $f' = -0.384$ → 2.923
- Closing in on 3, with η untouched

Table 3 on course.bioinfo.se/session01.html.

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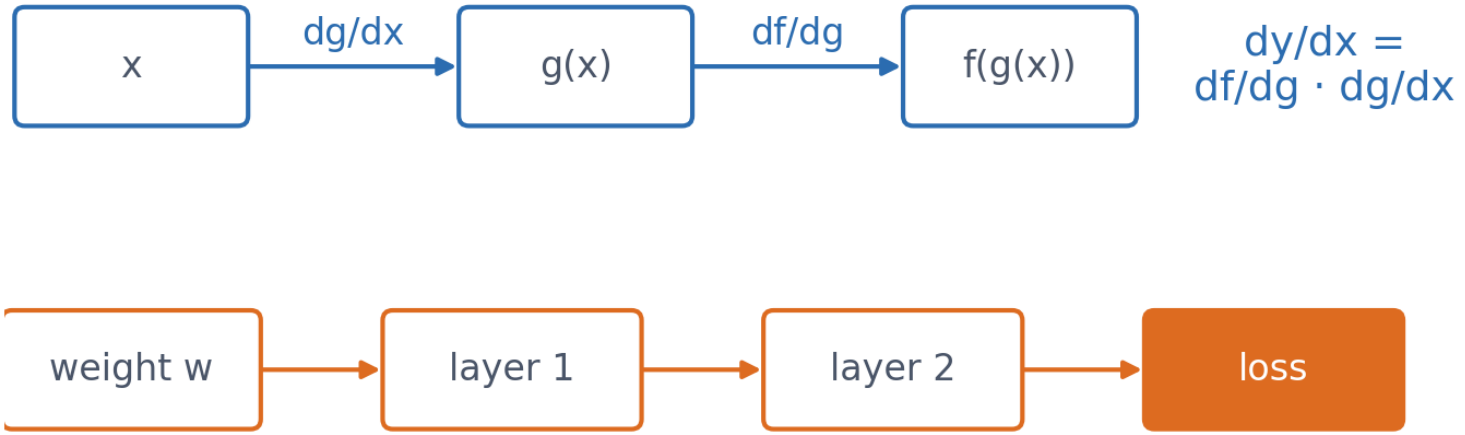
From one derivative to a million

- Here: one variable, one derivative
- **In a real model: every weight updated at once**
- The gradient vector replaces the single f′
- **Same rule, one subtraction per weight**
- You will see this loop from Session 7 onwards

See course.bioinfo.se/session01.html, 'Gradient descent'.

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The chain rule: multiply the derivatives along the chain

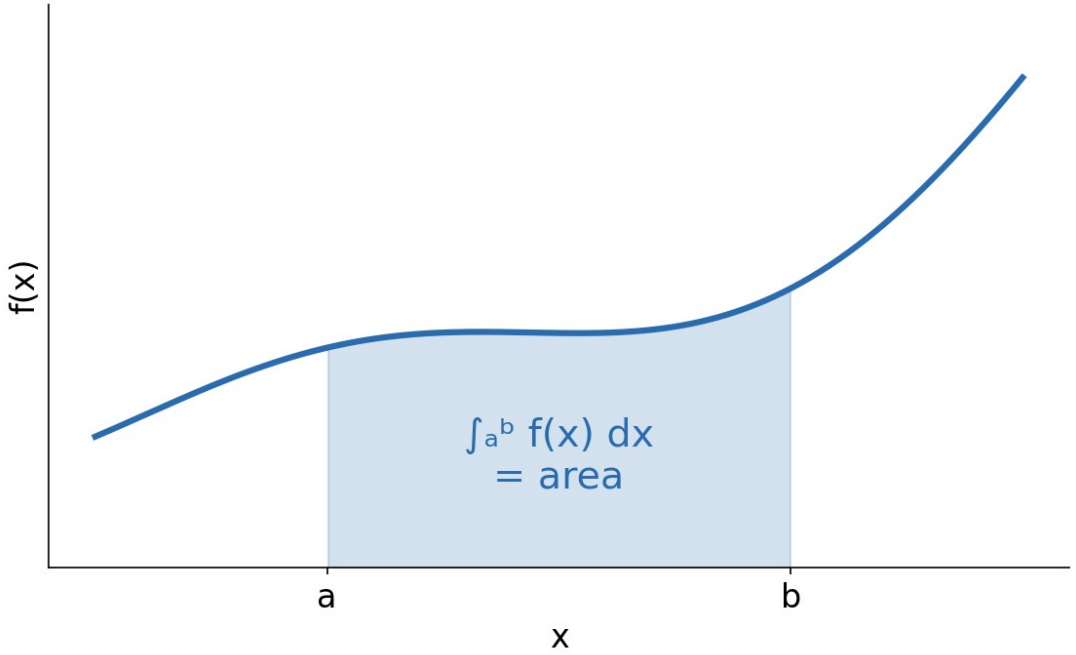


For y = f(g(x)), dy/dx = df/dg · dg/dx. A network's output is a function of a function of a function of its weights, so the chain rule gives the gradient of every weight.

See course.bioinfo.se/session01.html, 'The chain rule'; backpropagation in Session 9.

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An integral is the area under the curve



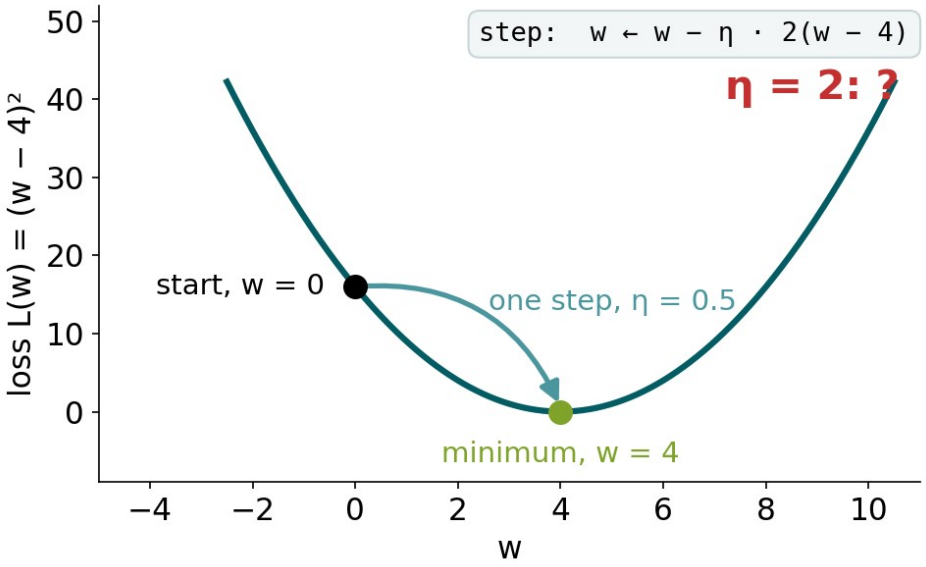
- Differentiation run backwards
- **AUROC is an area (Session 8)**
- Optional box on the page

See course.bioinfo.se/session01.html, 'Optional: integrals, briefly'.

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What happens with η = 2?

$L(w) = (w - 4)^2$, start at $w = 0$. With $\eta = 0.5$, one step lands exactly on the minimum.



A Lands on the minimum in one step

B Overshoots a little, then settles at w = 4

C Overshoots further each step, on alternating sides

D Overshoots once to w = 8 and stays there

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

SciLifeLab

What happens with η = 2?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

SciLifeLab

What happens with η = 2?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session01-discussion-2.ipynb

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What happens with η = 2?

Answer again

Same question, after the discussion.

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What happens with η = 2?

Results, and who explains

Both rounds side by side.

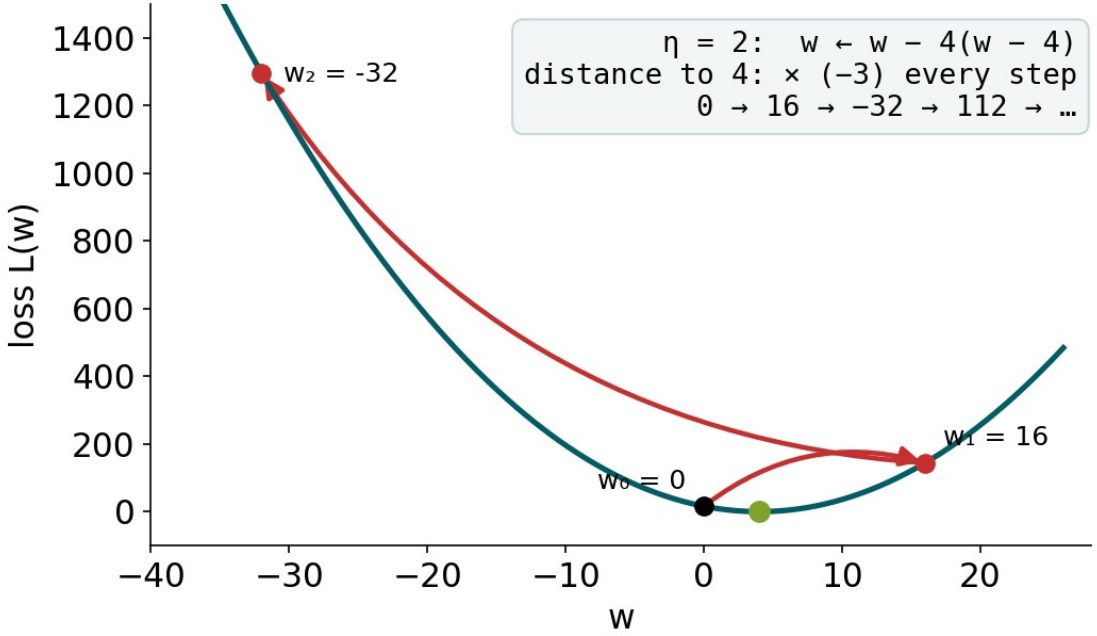
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Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 2 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session01-discussion-2.ipynb

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The answer: η = 2 moves away from the minimum



Each step multiplies the distance to the minimum by 1 − 2η. With η = 0.5 that is 0: one step lands on w = 4. With η = 2 it is −3: every step lands three times further away, on the other side. Too large a learning rate diverges.

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BLOCK 2 DONE

Break

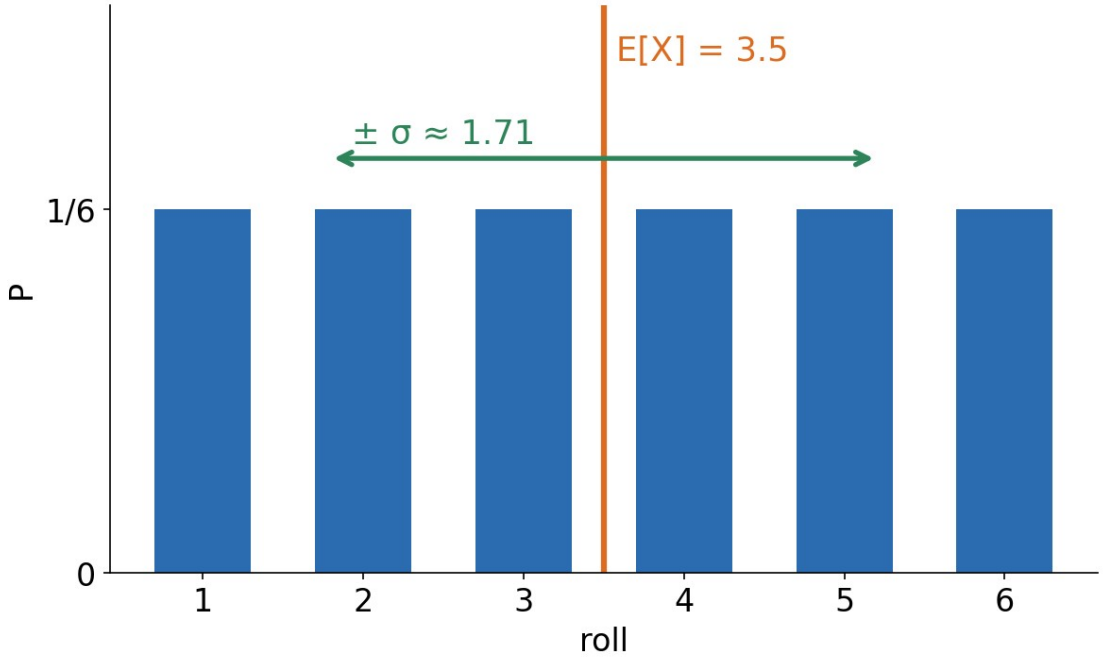
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BLOCK 3 · ABOUT 25 MIN

Probability and statistics

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Expectation and spread

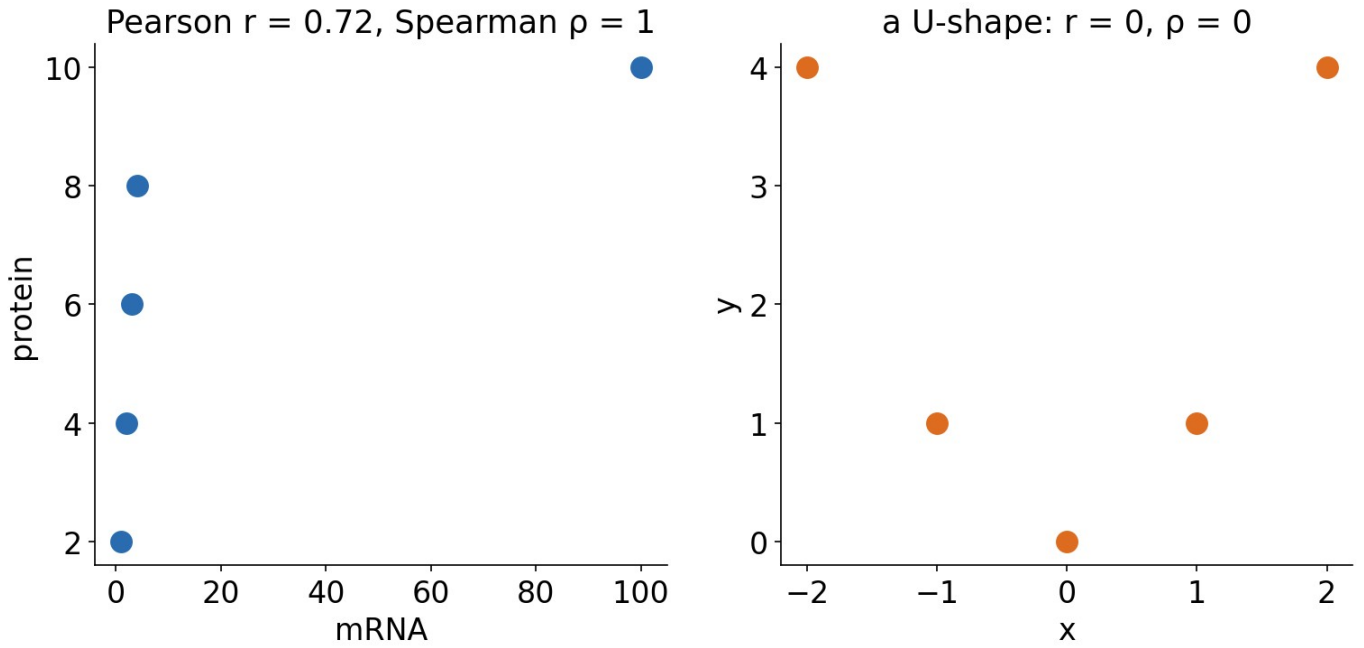


- **$E[X] = \Sigma\, x_i\, P(x_i)$**
- $Var(X) = E[(X - \mu)^2]$
- $\sigma = \sqrt{Var(X)}$

See course.bioinfo.se/session01.html, 'A short recap: expectation and variance'.

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Pearson and Spearman: a line, or a rising trend?

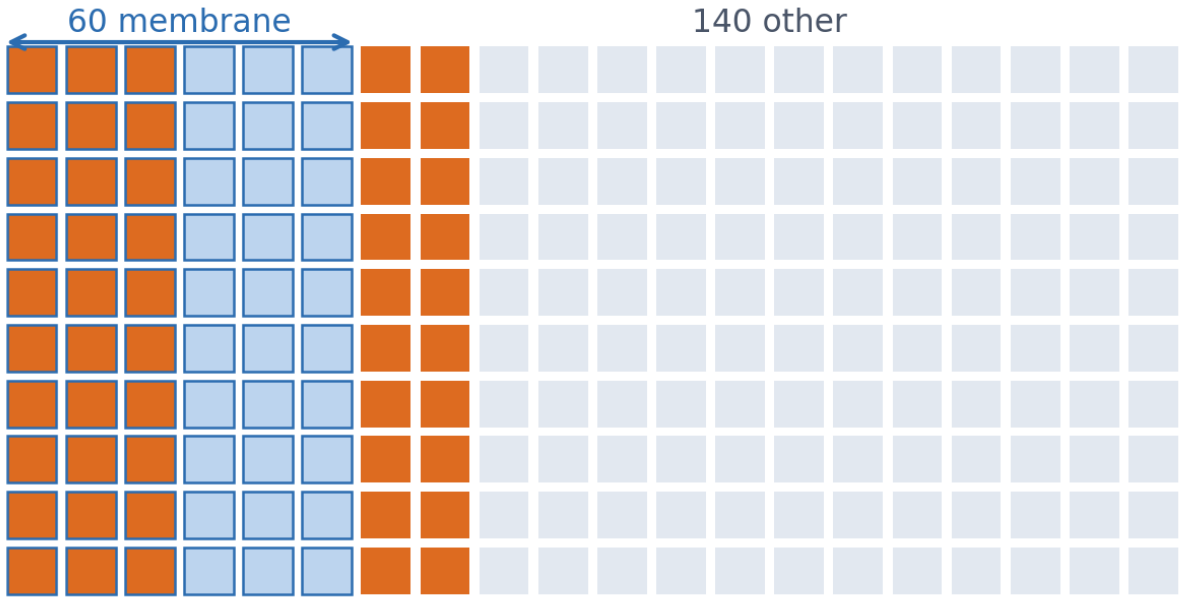


Left: every increase in mRNA goes with an increase in protein, so Spearman's ρ = 1, but one extreme value pulls Pearson's r down to 0.72. Right: both are 0 for a U-shape. Plot first.

See course.bioinfo.se/session01.html, 'Correlation: Pearson and Spearman'.

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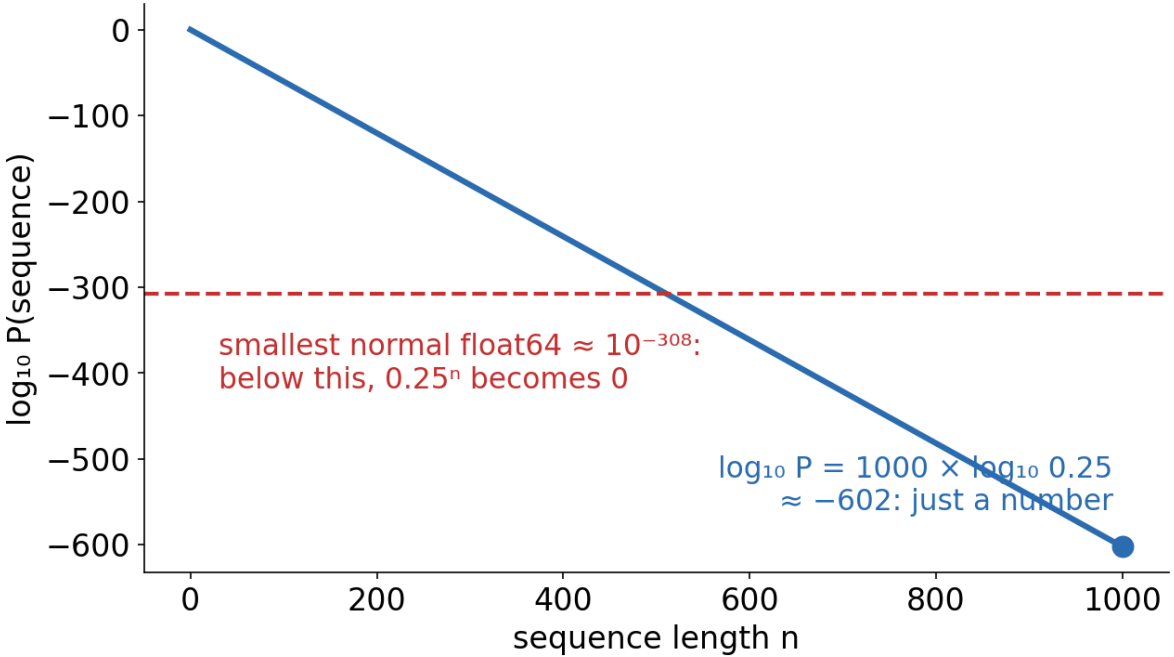
P(A | B): restrict to the cases where B holds



200 proteins: 60 membrane proteins, 30 of them with a signal peptide, and 20 of the other 140 with one too. The direction matters: 0.5 is not 0.6.

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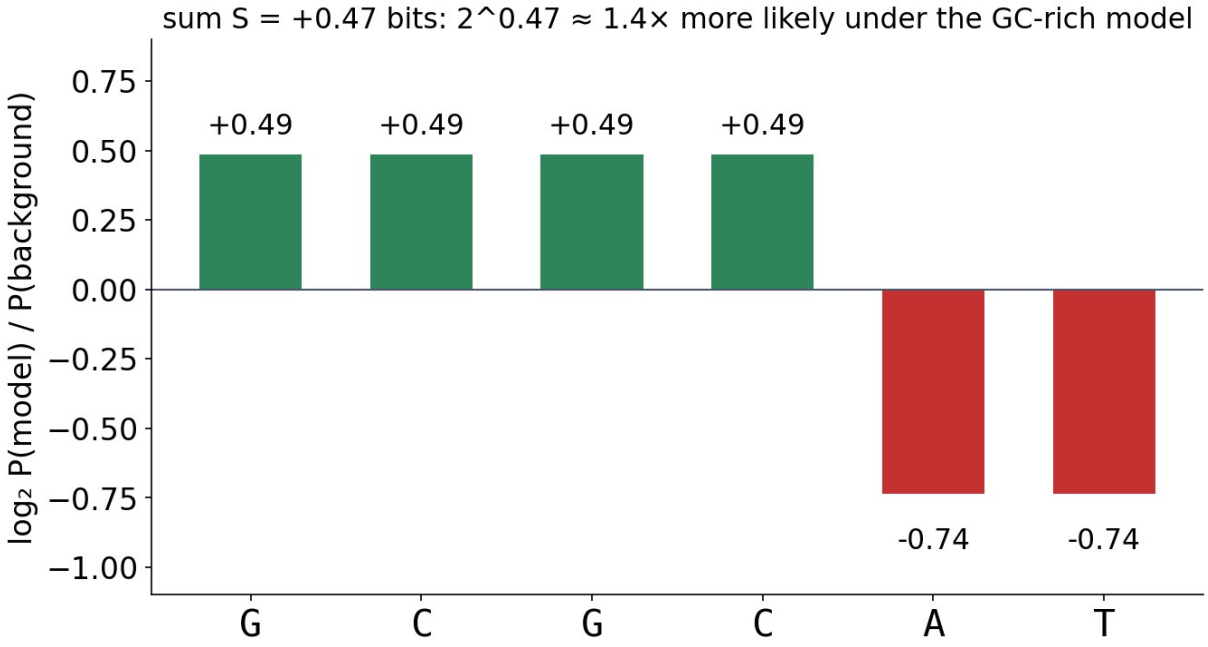
Why bioinformatics works with logs



- **$0.25^{1000} \approx 10^{-602}$ rounds to 0**
- $\log(xy) = \log x + \log y$
- Products become sums

See course.bioinfo.se/session01.html, 'Log probabilities and log-odds'.

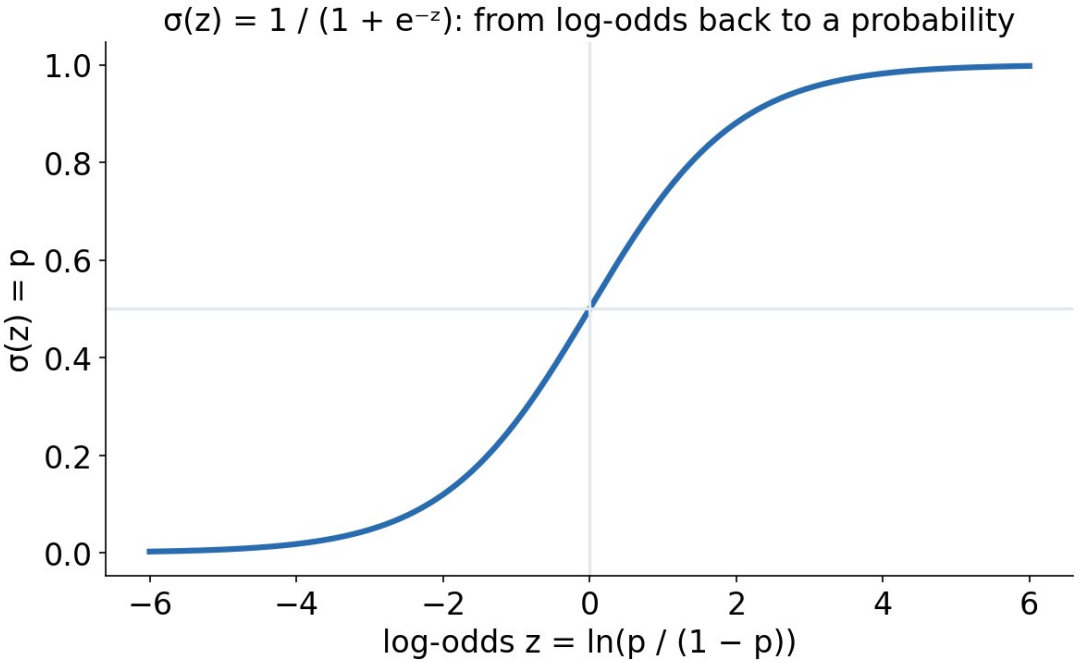
Log-odds: model against background, base by base



- **Add per-base log ratios**
- Positive: the model wins
- BLOSUM, PSSMs (Sessions 4-5)

See course.bioinfo.se/session01.html, 'Log probabilities and log-odds'.

The sigmoid: from log-odds back to a probability

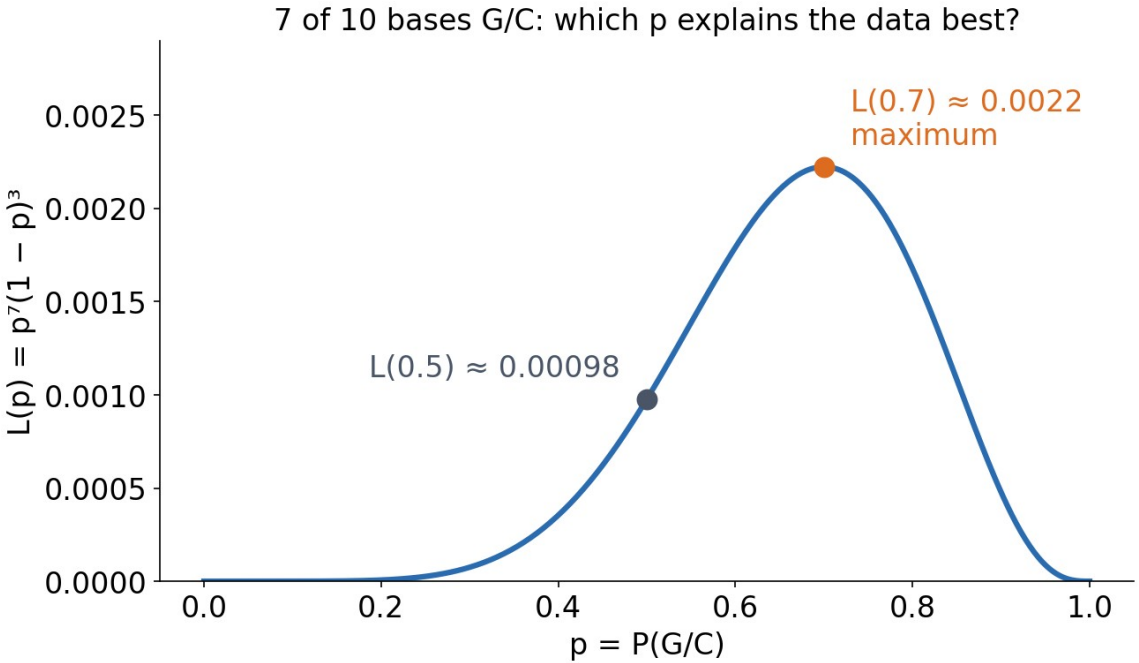


- **σ(ln(p / (1 − p))) = p**
- Logistic regression (Session 8)
- A calibrated probability only if trained for it

See course.bioinfo.se/session01.html, 'Log probabilities and log-odds'.

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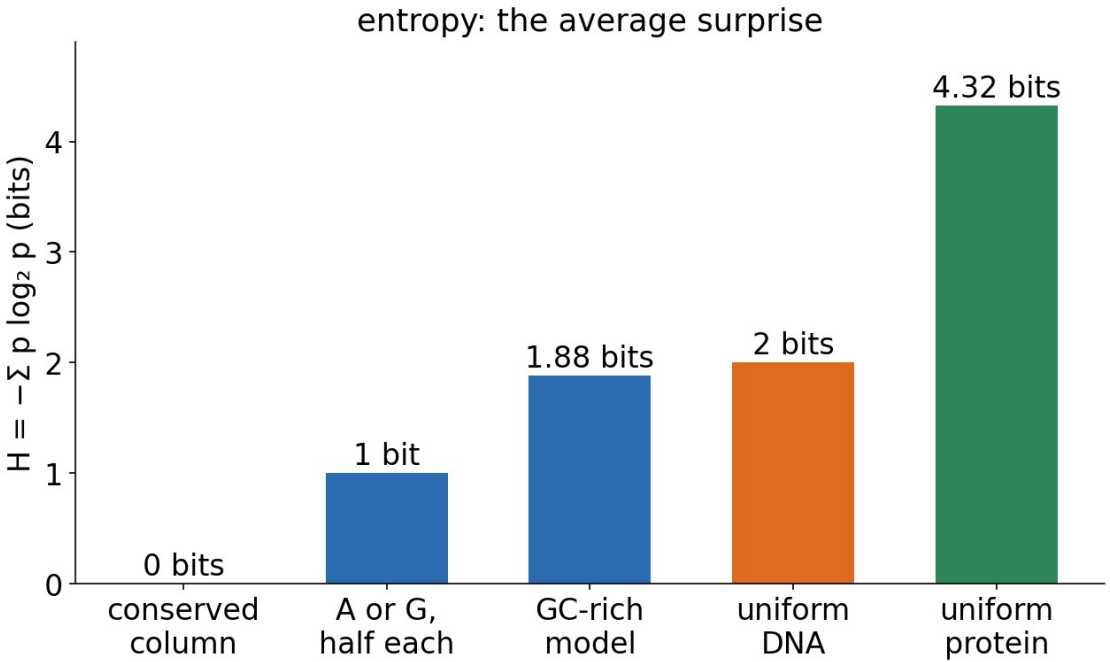
Likelihood: which parameters explain the data?



- **L(θ) = P(data | θ)**
- L(0.7) ≈ 2.3 × L(0.5)
- **Maximum likelihood: how models are trained**

See course.bioinfo.se/session01.html, 'Likelihood'.

Entropy: average surprise

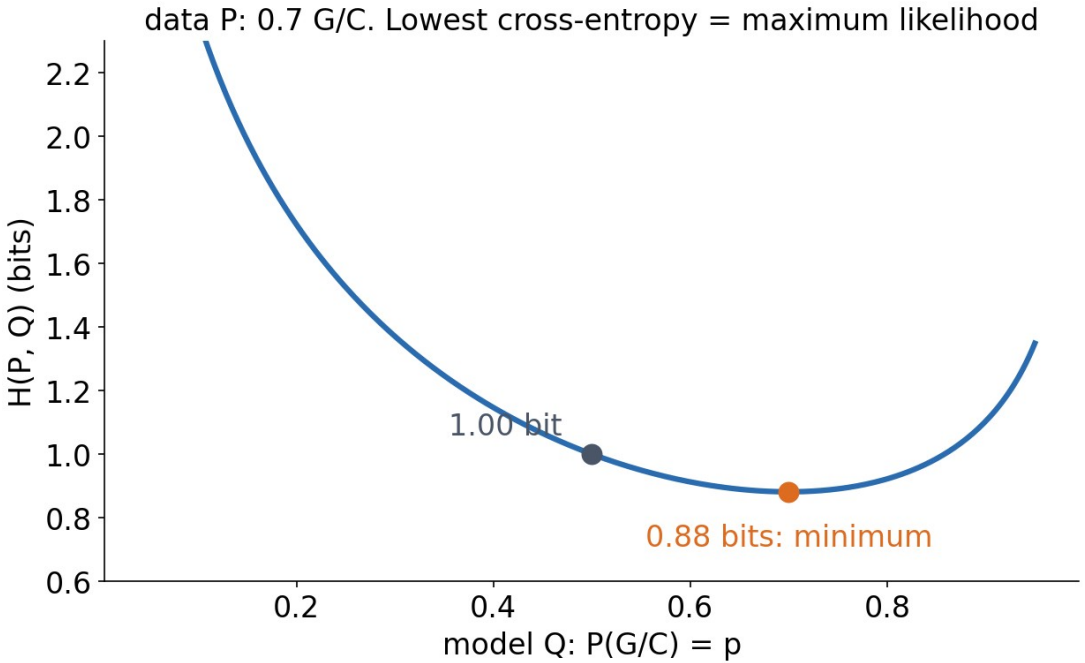


- **Surprise: $-\log_2 p$ bits**
- $H = -\Sigma\, p \log_2 p$
- Max − H = a logo's stack height (Session 4)

See course.bioinfo.se/session01.html, Table 1.3.

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Cross-entropy is maximum likelihood in disguise

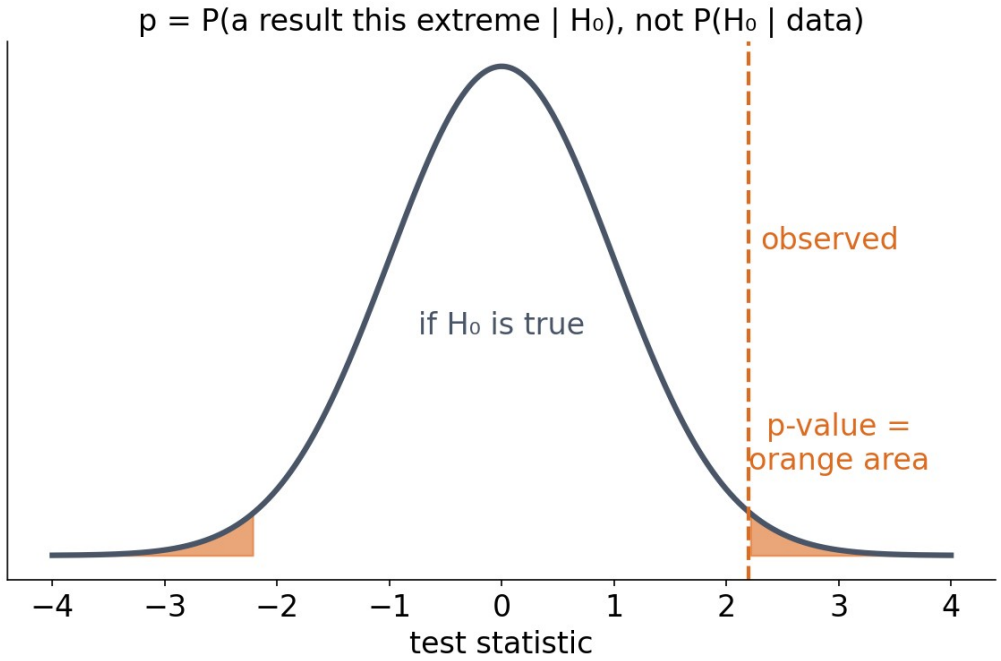


- **$H(P, Q) = -\Sigma P \log_2 Q$**
- Never below H(P)
- **The classifier loss (Sessions 8-9)**

See course.bioinfo.se/session01.html, 'Entropy: average surprise'.

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A p-value is a tail area under the null

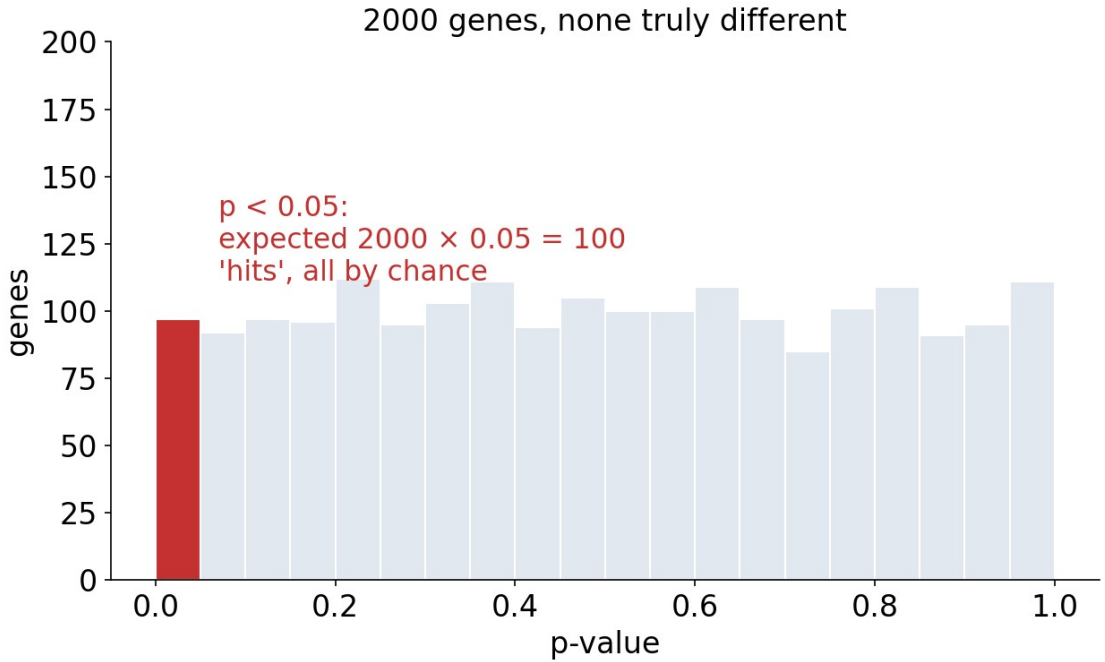


- **P(data this extreme | H_0)**
- **Not P(H_0 | data)**
- 0.05 is a convention, not a law

See course.bioinfo.se/session01.html, 'p-values and hypothesis testing'.

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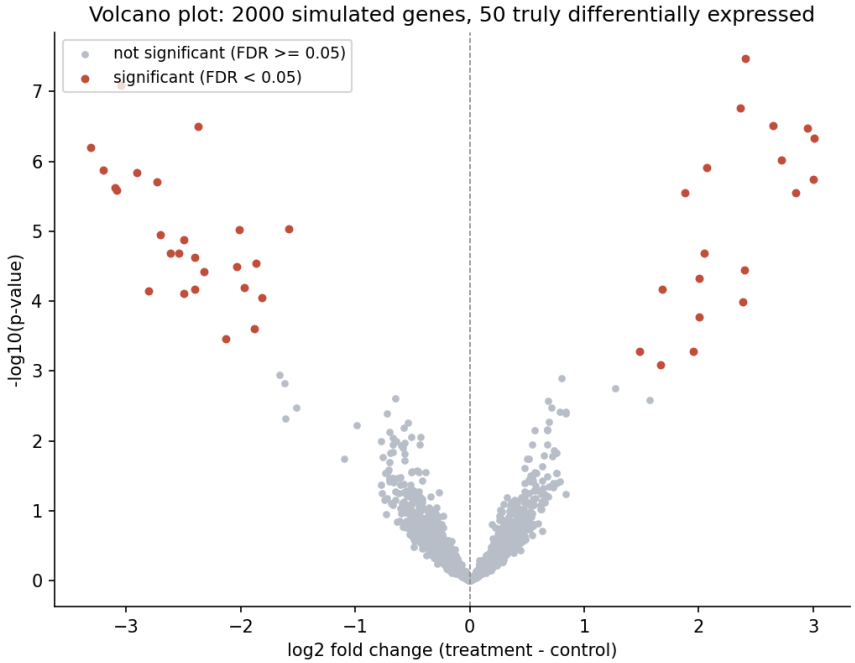
Test 2000 genes and chance alone gives about 100 hits



- **Null p-values are uniform**
- Bonferroni: often too strict
- **FDR: Benjamini-Hochberg**

See course.bioinfo.se/session01.html, 'The multiple testing problem'. The histogram is one simulation.

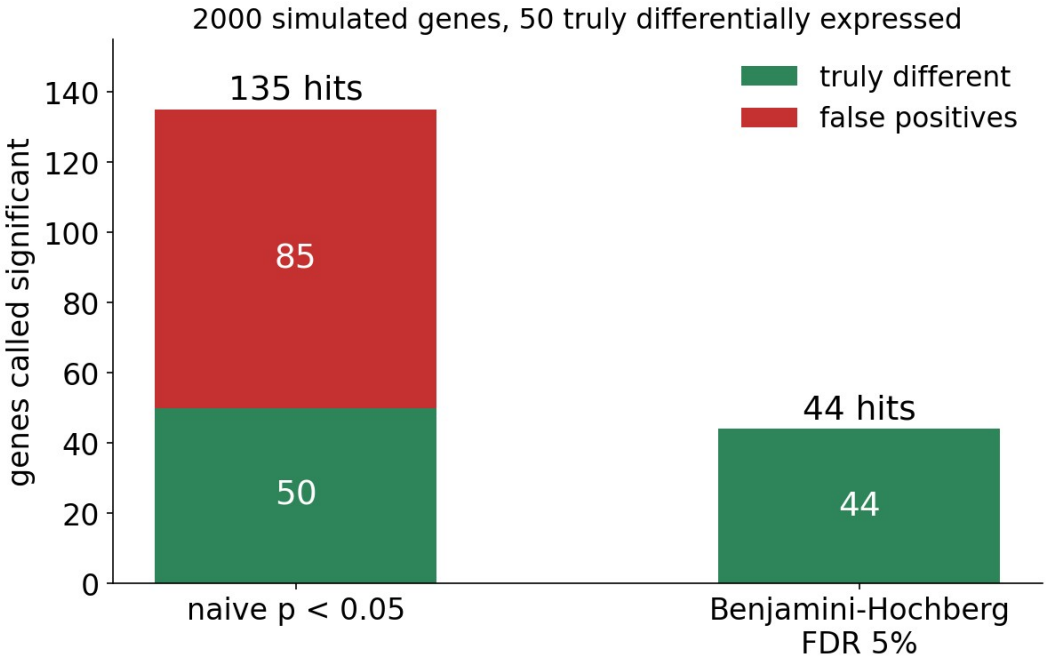
RNA-seq: the volcano plot



- x: $\log_2$ fold change (effect)
- y: $-\log_{10}$ p (evidence)
- **Interesting genes: far out and high up**

See course.bioinfo.se/session01.html, 'Worked example: RNA-seq differential expression and the volcano plot'.

Naive threshold against FDR control

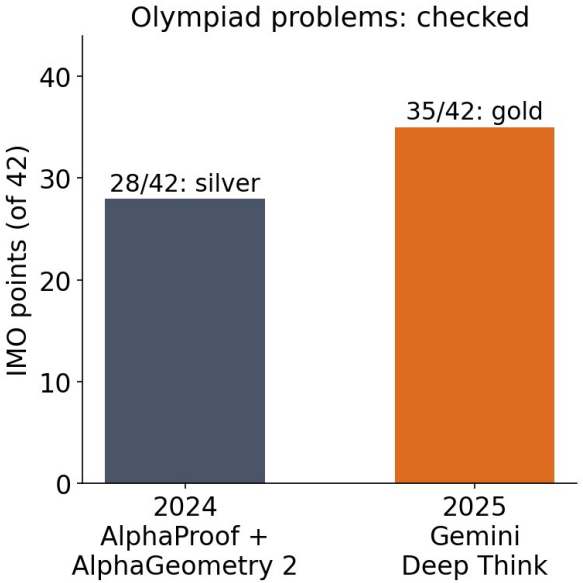


- **p < 0.05: 135 hits, 85 false**
- **BH at 5%: 44 hits, all real**
- 5% FDR does not promise zero

See course.bioinfo.se/session01.html, 'Worked example: RNA-seq differential expression and the volcano plot'.

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AI is doing mathematics: checked results and claims



Open research problems: claimed

372 results on open problems, released by OpenAI (2026)

~3 h of compute per result, on average

? how many are new, correct and important: being checked

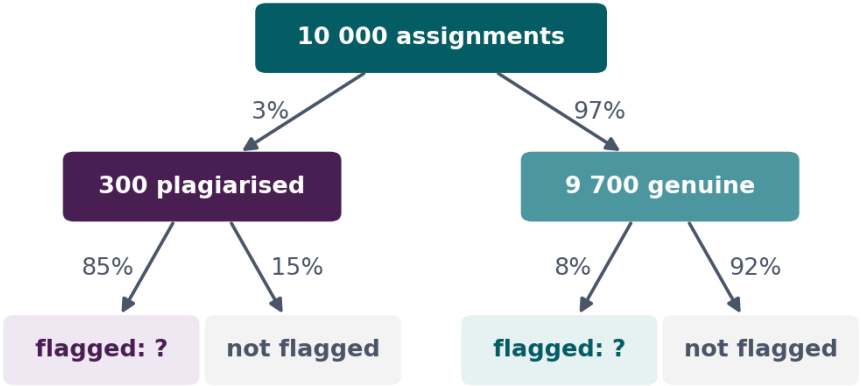
- 2024: olympiad silver; 2025: gold
- 2026: 372 claimed results on open problems
- Correct, new and important? Being checked
- **If an AI can do the maths, why learn it?**

DeepMind blog, 25 Jul 2024 (AlphaProof, AlphaGeometry 2) and Jul 2025 (Gemini Deep Think); The Decoder on OpenAI's release of 372 results (2026). https://deepmind.google/discover/blog/ai-solves-imo-problems-at-silver-medal-level/

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How likely is a flagged assignment plagiarised?

An automatic detector flags 85% of plagiarised and wrongly flags 8% of genuine assignments. 3% of all assignments are plagiarised. Your assignment is flagged.



Your assignment is flagged. Which flagged box is it more likely to be in?

A About 85%

B About 92%

C About 25%

D About 3%

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

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How likely is a flagged assignment plagiarised?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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How likely is a flagged assignment plagiarised?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session01-discussion-3.ipynb

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How likely is a flagged assignment plagiarised?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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How likely is a flagged assignment plagiarised?

Results, and who explains

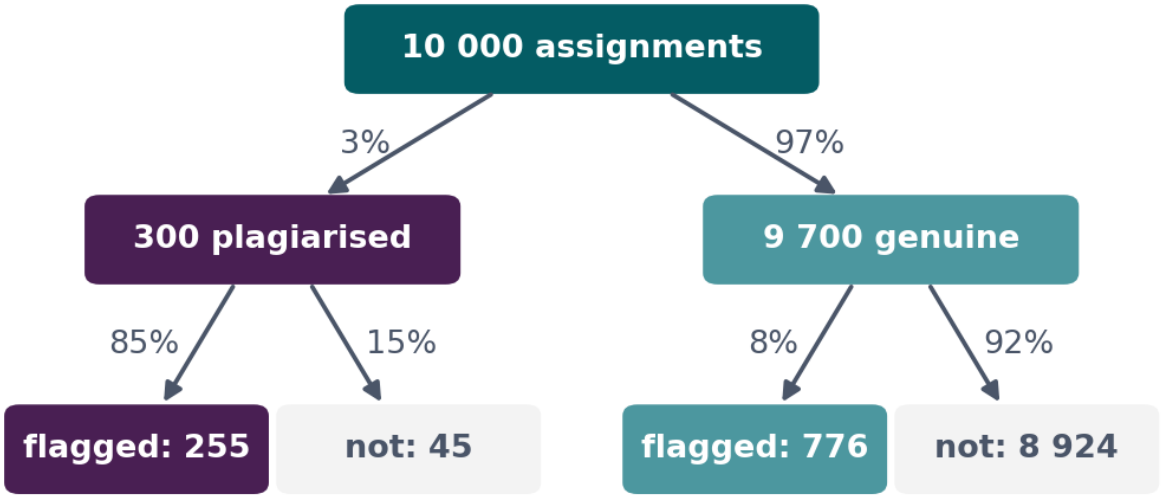
Both rounds side by side.

Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

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The answer: about 25%



P(plagiarised | flagged) = 255 / (255 + 776) = 255 / 1 031 ≈ 25%

Plagiarism is rare, so the 8% false flags among the many genuine assignments (776) outnumber the true flags among the few plagiarised ones (255). P(flagged | plagiarised) = 85%, but P(plagiarised | flagged) is only about 25%: the direction of conditioning matters.

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Today's maths, and where it comes back

Vectors and matrices	Calculus	Probability and statistics
profile scores (S5)	gradient descent (S7)	log-odds: BLOSUM, PSSM (S4-5)
perceptron (S7)	backpropagation (S9)	cross-entropy loss (S8-9)
every layer (S7-12)		maximum likelihood: training

today's maths, and where it comes back

The dot product is a profile score and a perceptron. The gradient and the chain rule train every network. Logs give log-odds scores, and cross-entropy is maximum likelihood.

See course.bioinfo.se/session01.html, 'What's next'.

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In up to three words: what was the hardest part of today?

Vectors and matrices

$x \cdot w$, $\|w\|$, Xw

Derivatives and gradient descent

$\partial L/\partial w$, ∇L, η

Probability and statistics

$P(A|B)$, log-odds, p

Answer now

Scan the code, or go to
quiz.bioinfo.se/live

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The hardest part of today

What the room found hardest

Show the results ▸ quiz.bioinfo.se/live/report

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Feedback on today

The Canvas survey

This session has no lab, so tell us now what to improve: "Sessions 0 and 1: feedback" in Canvas

Concrete suggestions: too long, missing background, errors, unclear slides

Takes about five minutes; Canvas records that you submitted it, not what you wrote

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Session 2 next

Every idea today written in code: Python, NumPy and PyTorch
The dot product becomes one line of PyTorch
Before Session 2: the Setup page (VS Code and Colab)